

## Original article

## Effects of Mediterranean climate annual variability on stream biological quality assessment using macroinvertebrate communities

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## ABSTRACT

Data on macroinvertebrates of selected reference sites were compiled from a long-term monitoring programme carried out in the Mediterranean Catalan Basins (NE Spain) that permitted analysis for nine years, from 1996 to 2004, using a homogeneous data collection procedure. This study aims to analyse the differences in composition and structure of macroinvertebrate communities at family level in five Mediterranean river types, and the values of biological quality metrics (IBMWP and IASPT indices, taxon richness and EPT) in reference conditions. Also differences between seasons (spring vs. summer) and between dry and wet periods were analysed. The dry and wet periods were determined using the Standardised Precipitation Index (SPI). A total of 29 reference sites were selected out of 184 sampling sites analysed, and 171 reference samples were available (from 1996 to 2004), of which 88 were sampled in dry periods, whereas 83 correspond to wet periods. Differences on community composition at family level were appreciated, clustering the rivers in three different groups: (1) rivers with a continuous flow regime located in siliceous zones; (2) rivers with a continuous flow regime located in calcareous zones; and (3) temporary rivers regardless of geology. Moreover, our results explain that the characteristics of hydrological periods (dry and wet) characterize the differences between communities better than just the season. The analysis of four biological quality metrics reveals clear differences between values obtained from dry and wet periods concerning taxon richness, EPT values and IBMWP biological indices, whereas the IASPT index does not show significant differences. The median taxonomic richness in wet periods is 32 macroinvertebrate families per sample while in dry periods this value falls to 22. Reference values of IBMWP index, the total number of taxa, and EPT metric are different between dry and wet periods in spring samples, while these differences are not relevant for IASPT index except for temporary streams. Hydrological specific conditions should additionally be considered in order to better calculate biological reference conditions, and to properly apply biological quality metrics used to establish the ecological status in Mediterranean rivers, especially in temporary ones. The use of the dry–wet period classification according to the climate characteristic results is a more accurate application of the Water Framework Directive in Mediterranean rivers. Implications of future climate change should be also considered from our results.

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## 1. Introduction

The Water Framework Directive (WFD) (European Commission, 2000) requires the EU Member States to define reference conditions in order to: (1) characterize the biological communities, (2) define the quality class boundaries using biological quality metrics, and (3) identify and consequently correct impacts from human activities which significantly affect the ecological status of water bodies (Wallin et al., 2003; Noges et al., 2009). That is why reference

sites need to be previously selected in each river type using areas where there is no human alteration or, if present, does not cause relevant ecological changes in the aquatic ecosystems (European Commission, 2003; Nijboer et al., 2004). Subsequently, reference conditions must be established by calculating the biological quality metrics in a significant number of reference sites within each river type. Moreover, reference conditions should be homogeneous within each river type, and be capable of differentiating river types for a correct ecological status assessment.

Methods based on biological community composition have occasionally been used to define river types (e.g., Heino et al., 2003; Dodkins et al., 2005), although biological assemblages can be heavily affected by human activities. Rivers may be mistakenly

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classified using biological communities when human pressures are present, hence the use of environmental descriptors is suggested for river classification purposes by the WFD. Thus, in most European countries, river types have been mainly defined using environmental descriptors (e.g., Dawson et al., 2002; Verdonshot and Nijboer, 2004; Munné and Prat, 2004; Ferréol et al., 2005), following the WFD requirements and, therefore, the number of river types defined tends to increase according to the high environmental heterogeneity. Nevertheless, it is not clear that biological assemblages are consistent with river types defined using environmental descriptors, and biological data should additionally be used to define homogeneous river types and their reference conditions (e.g., Sánchez-Montoya et al., 2007). Moreover, it is also well known that Mediterranean rivers are characterized by a high temporal variability, combining dry and wet periods and this makes the characterization of these rivers even more difficult. In Mediterranean streams, the hydrological regime is a key element that determines community composition (Poff, 1996; Bonada et al., 2007a), and its response to the interannual and seasonal hydrological variability (Gasith and Resh, 1999; Williams, 1996; Lake, 2003). Numerous studies have revealed the peculiarities of biological communities in Mediterranean and temporary streams (e.g., Boulton and Lake, 1992; Williams, 2006), where changes in the composition of the invertebrate community are found over time related to flow regime variation (Ortega et al., 1991; Blinn et al., 1995; Gallardo-Mayenco et al., 1995; Acuña et al., 2005; Bonada et al., 2007a). Thus, reference conditions could change between seasons, between dry and wet periods, and after hydrological events in the same river type, which complicates the biological quality assessment on Mediterranean rivers.

However, changes over time of ecological quality assessment and reference conditions have been scarcely investigated (e.g., Rose et al., 2008; Death et al., 2009), and their consequences for water management are virtually unknown. This study attempts to analyse the differences in composition and structure of macroinvertebrate communities at family level in various Mediterranean river types, and their effects on values of biological quality metrics in reference conditions. Our main objective is to analyse the differences among river types, and between dry and wet periods, using a data set from nine consecutive years (1996–2004). We analyse the performance of different metrics used for river quality monitoring in Mediterranean streams (IBMWP and IASPT indices, taxon richness and EPT metrics) under different climatic circumstances (dry and wet periods), and the variation of quality class boundaries between very good and good, and between good and moderate ecological status, according to the WFD requirements.

## 2. Material and methods

### 2.1. Study area and sampling sites

A total of 184 sampling sites located in the Catalan Mediterranean basins (NE Spain) (Fig. 1) were analysed in order to find reference sites. The basins analysed have a total area of 16,438 km<sup>2</sup> and an average annual precipitation ranging from 350 mm, in the sites located in semi-arid zones, up to 1400 mm in headwater sites located in mountain zones. The average annual temperature varies from 7 °C to 15 °C, and the flow discharge from a scarce m<sup>3</sup> s<sup>-1</sup>, or completely dry in some periods, up to 30 m<sup>3</sup> s<sup>-1</sup> with a high annual variability. All sampling sites were located in the Mediterranean climate region (Illies and Botosaneanu, 1963), and were classified into five different river types: (1b, 2a, 2b, 3a and 3c) out of ten river types defined in the Catalan Mediterranean basins using environmental descriptors according to the WFD requirements (see Munné and Prat, 2004 for details) (Table 1).



Fig. 1. Sampling sites analysed over a period of nine consecutive years (1996–2004) in Mediterranean Catalan basins are shown using grey circles. The selected reference sites are distinguished by a black squares.

### 2.2. Selection of reference sites

In selecting reference sites, several works previously carried out in the Spanish Mediterranean rivers were considered (i.e., CHE, 1999; Bonada et al., 2004; Munné and Prat, 2009). Also, the criteria provided by the European Commission (European Commission, 2003; Heiskanen et al., 2004), and the criteria agreed during the intercalibration exercise carried out in Mediterranean river types (Mediterranean Geographic Intercalibration Group–MedGIG) (European Commission, 2007) were taken into account. Using all these documents, and according to the data available, a total of 42 criteria for selecting reference sites were established at three different scales: human pressures at basin, reach (river segment and its floodplain area), and sampling site (Table 2). A Geographic Information System (ARCGIS software) was used to analyse pressures considering aerial photos, maps, and land use layers, together with measured and observed data in the field (ACA–Agència Catalana de l'Aigua, 2005). The nature of river beds and their banks, and the chemical water quality at site scale was considered to be an inviolable reference criterion in order to select reference sites (Bonada et al., 2004; European Commission, 2007), and consequently an assessment *in situ* of the hydromorphological characteristics and water chemical analysis was carried out. Two thresholds were defined in order to select the reference sites, on the one hand the selection threshold at which sites were considered true reference sites and, on the other, an exclusion threshold based on key elements from where sites were rejected (Table 2).

### 2.3. Characterization of dry and wet periods

The dry and wet periods were determined using the Standardised Precipitation Index (SPI) (McKee et al., 1993). The SPI was developed at Colorado State University, and is widely used to analyse and select drought periods in semi-arid areas (e.g., Edwards and McKee, 1997; Tsakiris and Vangelis, 2004). SPI is basically a representation of rainfall of the period in units of standard deviation. Positive values indicate greater than median rainfall, whereas negative values indicate less than median rainfall. It assumes that the series of precipitations received by a basin follow a normal distribution.

**Table 1**  
Type and number of reference sites selected using Table 2 criteria sites for each river. Samples collected in spring and summer from the ECOSTRIMED data base, and grouped for dry and wet periods according to the SPI values are shown.

| River type code | River type name                                                                                    | Major distinguishing features                                                                                                                                                                                                                                                                                                        | Total sampling sites | Reference sampling | Season           | Number of reference samples(W96–20M) |            |
|-----------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------|------------------|--------------------------------------|------------|
|                 |                                                                                                    |                                                                                                                                                                                                                                                                                                                                      |                      |                    |                  | Dry period                           | Wet period |
| 1a              | Medium-size rivers with a continuous flow regime located in a siliceous wet-high mountains         | Medium to low drainage area (50–300 km <sup>2</sup> ). High altitude (500–800 m a.s.l.). Moderate to low annual discharge (2–5 m <sup>3</sup> s <sup>−1</sup> ). High percentage of siliceous surface geology in the catchment (>80%). Low annual average temperature (9 °C). High annual rainfall (1000–1400 mm)                    | 5                    | 0                  | –                |                                      |            |
| 1b              | Medium-size rivers with a continuous flow regime located in a calcareous wet-high mountains        | Medium to low drainage area (50–300 km <sup>2</sup> ). High altitude (400–700 m a.s.l.). Moderately to low annual discharge (2–6 m <sup>3</sup> s <sup>−1</sup> ). High percentage of calcareous surface geology in the catchment (>80%). Moderate to low annual average temperature (11 °C). Moderate annual rainfall (900–1000 mm) | 6                    | 3                  | Spring           | 6                                    | 5          |
| 2a              | Small rivers with a continuous flow regime located in a siliceous Mediterranean climate mountains  | Low drainage area (10–100 km <sup>2</sup> ). Mid-altitude (300–600 m a.s.l.). Low annual discharge (<2 m <sup>3</sup> s <sup>−1</sup> ). High percentage of siliceous surface (>60%). High annual average temperature (13–14 °C). Moderate annual rainfall (800–900 mm).                                                             | 39                   | 13                 | Summer<br>Spring | 5<br>12                              | 4<br>19    |
| 2b              | Small rivers with a continuous flow regime located in a calcareous Mediterranean climate mountains | Low drainage area (10–100 km <sup>2</sup> ). Mid-altitude (300–600 m a.s.l.). Low annual discharge (<2 m <sup>3</sup> s <sup>−1</sup> ). High percentage of calcareous surface geology in the catchment (>80%). Moderate to high annual average temperature (10–13 °C). Moderate to high annual rainfall (800–1100 mm)               | 13                   | 5                  | Summer<br>Spring | 13<br>6                              | 14<br>7    |
| 2c              | Mediterranean mountain rivers with high discharge                                                  | Moderate drainage area (100–1000 km <sup>2</sup> ). Mid-altitude (300–600 m a.s.l.). Moderate to high annual discharge (3–20 m <sup>3</sup> s <sup>−1</sup> ). Moderate to high annual average temperature (10–13 °C). Moderate to high annual rainfall in the catchment (800–1100 mm).                                              | 21                   | 0                  | Summer           | 5                                    | 7          |
| 3a              | Small rivers with a temporary flow regime located in calcareous lowlands                           | Low drainage area (10–100 km <sup>2</sup> ). Low-altitude (<300 m a.s.l.). Low annual discharge (<1 m <sup>3</sup> s <sup>−1</sup> ). High flow variability (temporary rivers). High annual average temperature (13–15 °C). Low annual rainfall (600–800 mm)                                                                         | 66                   | 6                  | Spring           | 14                                   | 13         |
| 3b              | Small rivers with a temporary flow regime located in a siliceous dry Mediterranean climate zone    | Low drainage area (10–100 km <sup>2</sup> ). Low-altitude (<300 m a.s.l.). Low annual discharge (<1 m <sup>3</sup> s <sup>−1</sup> ). High flow regime variability (temporary rivers). High percentage of siliceous surface geology (>80%). High annual average temperature (13–14 °C). Low annual rainfall (700–750 mm)             | 5                    | 0                  | Summer<br>–      | 13<br>–                              | 9          |
| 3c              | Small karst feed rivers with a temporary flow regime                                               | Low drainage area (10–100 km <sup>2</sup> ). Low-altitude (<300 m a.s.l.). Low annual discharge (<2 m <sup>3</sup> s <sup>−1</sup> ). Low flow variability. High percentage of calcareous surface geology (>80%). High annual average temperature (13–14 °C). Low annual rainfall (500–600 mm)                                       | 14                   | 2                  | Spring           | 5                                    | 5          |
| 4a              | Mediterranean large rivers                                                                         | High drainage area (1000–5000 km <sup>2</sup> ) High annual discharge (10–40 m <sup>3</sup> s <sup>−1</sup> )                                                                                                                                                                                                                        | 8                    | 0                  | Summer<br>–      | 4<br>–                               | 5          |
| 5a              | Small coastal ephemeral rivers                                                                     | Small streams located along of the littoral (<200 m a.s.l.). Small drainage area (<250 km <sup>2</sup> ). Ephemeral flow regime (>120 dry days per year)                                                                                                                                                                             | 7                    | 0                  |                  | –                                    |            |

**Table 2**

Criteria used for selecting reference sites. The source of data is shown for each criterion: (S) already existing geographic and cartographic information systems, (E) measured data using aerial photos and maps, (C) data measured or observed in the field and/or from laboratory studies, and (T) data from technical reports.

| Scale               | Type of human pressure                                                                                                                                                                                                                                                                       | Criteria                                                                                                      | Selection threshold | Exclusion threshold | Source of data |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------|---------------------|----------------|
| Basin level         | Diffuse contamination Human activity in the basin                                                                                                                                                                                                                                            | Urban and industrial land uses                                                                                | <0.4%               | >2%                 | S              |
|                     |                                                                                                                                                                                                                                                                                              | Superficial mining land uses                                                                                  | <0.4%               | >2%                 | S              |
|                     |                                                                                                                                                                                                                                                                                              | Intensive agriculture (vineyards and orchards)                                                                | <1%                 | >15%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Agriculture (Irrigated fields)                                                                                | <10%                | >15%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Intensive agriculture (cereal, etc.)                                                                          | <20%                | >50%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Non intensive agriculture (pastures, etc)                                                                     | <30%                | >50%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Forestry, tree plantations (non native species)                                                               | <30%                | >50%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Landfill site                                                                                                 | <0.4%               | >2%                 | S              |
|                     |                                                                                                                                                                                                                                                                                              | Cattle and pigs breeding (animal units/ha)                                                                    | <1.25               |                     | S              |
|                     |                                                                                                                                                                                                                                                                                              | Natural areas                                                                                                 | >70%                | <50%                | S              |
| River segment level | Point source pollution (sewage plants, non treated waste water discharges, etc.)                                                                                                                                                                                                             | Increase of organic matter (COD) in the river (absolute values of COD dumped (mg)/ river flow (L))            | <5 mg/L             |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              | Increase of total phosphorus (P) in the river (absolute values of P dumped (mg) / river flow (L))             | <1 mg/L             |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     | Morphological alteration: floodplain occupation<br>Width of floodplain: (Q 500 occupation or 50m for small streams (order 1–3), 150m for medium size rivers (order 4–5) and 300 m for larger rivers (order $\geq$ 6)                                                                         | Urban land uses                                                                                               | <0.4%               | >2%                 | S              |
|                     |                                                                                                                                                                                                                                                                                              | Extractive activities                                                                                         | <10%                | >20%                | E              |
|                     |                                                                                                                                                                                                                                                                                              | Intensive agriculture (vineyards and orchards)                                                                | Non-existent        |                     | S              |
|                     |                                                                                                                                                                                                                                                                                              | Irrigated fields                                                                                              | <10%                | >20%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Non intensive agriculture                                                                                     | <20%                | >30%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Fast-growth riparian forestry, tree plantations (non native species)                                          | <20%ACA1            |                     | S              |
|                     |                                                                                                                                                                                                                                                                                              | Urban land uses                                                                                               | <0.2%               | >0.8%               | S              |
|                     |                                                                                                                                                                                                                                                                                              | Extractive activities                                                                                         | <0.2%               | >0.8%               | E              |
|                     |                                                                                                                                                                                                                                                                                              | Intensive agriculture (vineyards and orchards)                                                                | Non-existent        |                     | S              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     | Morphological alteration: riparian corridors occupation. Width of riparian corridor: (Q50 (50 years flooding return) boundary (when the information is available) or: 30m for small streams (order 1–3), 50m for medium size rivers (order 4–5) and 100 m for larger rivers (order $\geq$ 6) | Irrigated fields                                                                                              | Non-existent        |                     | S              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     |                                                                                                                                                                                                                                                                                              | Non intensive agriculture                                                                                     | <10%                | >20%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Forestry, tree plantations (non native species)                                                               | <10%                |                     | E              |
|                     |                                                                                                                                                                                                                                                                                              | Cattle breeding or farming                                                                                    | Non-existent        | >20%                | C              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     |                                                                                                                                                                                                                                                                                              | Natural areas                                                                                                 | > 70%               | <50%                | S              |
|                     |                                                                                                                                                                                                                                                                                              | Number of dams or weirs per river length (Km)                                                                 | <0.5                |                     | S              |
|                     |                                                                                                                                                                                                                                                                                              | Dams which significantly modify the natural hydrological flow regime (flow regulation) located upstream       | Non-existent        |                     | T              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     | Artificial instream or bank structures and river profiles                                                                                                                                                                                                                                    | Artificial bank structures along the river segment                                                            | <10%                |                     | T              |
|                     |                                                                                                                                                                                                                                                                                              | River profiles along the river segment                                                                        | <20%                |                     | T              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     | Water abstraction                                                                                                                                                                                                                                                                            | Cumulative effect of water abstraction compared with the flow discharge                                       | <20%                |                     | T              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     | Invasive species                                                                                                                                                                                                                                                                             | Alien species (animals or plants) in stage of active colonisation, which are quantitatively predominant       | Non-existent        |                     | T              |
|                     |                                                                                                                                                                                                                                                                                              | Non indigenous species (animals or plants) recently introduced, not known to present a risk of being invasive | <3                  |                     | T              |
| Sampling site level | Flow regime alteration                                                                                                                                                                                                                                                                       | Significant water abstraction or diversion in the reach (by-passed section with small residual flow)          | Non-existent        |                     | C, T           |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     | Physical and chemical conditions (water quality)                                                                                                                                                                                                                                             | Ammonia: (N–NH <sub>4</sub> <sup>+</sup> ) mean of values                                                     | <0.2 mg/L           |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              | Ammonia: (N–NH <sub>4</sub> <sup>+</sup> ) maximum of values                                                  | <1 mg/L             |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              | Nitrate (N–NO <sub>3</sub> <sup>–</sup> ) mean of values                                                      | <10 mg/L            |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              | Nitrate: (N–NO <sub>3</sub> <sup>–</sup> ), maximum of values                                                 | <20 mg/L            |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              | Phosphate: (P–PO <sub>4</sub> <sup>–</sup> ) mean of values                                                   | <0.1 mg/L           |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              | Phosphate: (P–PO <sub>4</sub> <sup>–</sup> ) maximum of values                                                | <1 mg/L             |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              |                                                                                                               |                     |                     |                |
|                     | Morphological alteration: riparian habitat quality                                                                                                                                                                                                                                           | Riparian quality index “QBR” (Munné et al., 2003)                                                             | >75                 |                     | C              |
|                     |                                                                                                                                                                                                                                                                                              | Riparian quality index “QBR” (Section 4-channel alteration)                                                   | >20                 |                     | C              |

**Table 3**

SPI values for each basin analysed in spring and summer seasons, and for a period of 9 years (1996–2004). Spring includes March, April and May, while summer includes June, July and August. SPI values lower than  $-0.5$  are considered as dry periods and are highlighted using an asterisk. Analysed basins are shown in each column.

| SPI values for each basin |        |          |        |           |        |         |        |        |        |
|---------------------------|--------|----------|--------|-----------|--------|---------|--------|--------|--------|
| Year                      | Season | Francolí | Foix   | Llobregat | Besòs  | Tordera | Ter    | Fluvià | Muga   |
| 1996                      | Spring | −0.43    | −0.33  | −0.22     | 0.71   | 1.09    | −0.92* | −0.67* | −0.66* |
|                           | Summer | −0.47    | −0.48  | −0.44     | −0.43  | 1.66    | −1.72* | −0.82* | −0.47  |
| 1997                      | Spring | −0.96*   | −0.87* | −0.96*    | −0.52* | −0.52*  | −1.67* | −1.33* | −0.63* |
|                           | Summer | 0.29     | −0.08  | −0.24     | −0.16  | −0.12   | −0.46  | −0.16  | 0.90   |
| 1998                      | Spring | −0.72*   | −0.76* | −1.00*    | −0.55* | −0.36   | −0.65* | −0.95* | −0.42  |
|                           | Summer | −0.33    | −0.47  | −0.90*    | −1.15* | −0.58*  | −0.69* | −0.78* | −0.91* |
| 1999                      | Spring | −0.01    | −0.55* | −0.93*    | −1.51* | −0.94*  | −1.14* | −1.19* | −0.36  |
|                           | Summer | −0.32    | −0.21  | −0.08     | −0.70* | −0.81*  | −1.03* | −0.31  | 0.26   |
| 2000                      | Spring | 0.26     | 0.22   | −0.48     | 1.04*  | −0.28   | −1.17* | −1.11* | −0.50  |
|                           | Summer | −0.96*   | −1.39* | −1.12*    | −1.76* | −1.57*  | −2.20* | −2.15* | −1.38* |
| 2001                      | Spring | −0.80*   | −1.00* | −1.47*    | −1.44* | −1.39*  | −2.12* | −0.81* | −1.04* |
|                           | Summer | −0.58*   | −1.87* | −1.60*    | −1.26* | −1.27*  | −1.85* | −1.62* | −1.29* |
| 2002                      | Spring | 0.33     | 0.37   | 0.15      | 0.89   | 3.14    | 2.25   | −0.03  | 1.74*  |
|                           | Summer | −0.55*   | −0.41  | −0.14     | 0.43   | 0.11    | −1.18* | −0.07  | −0.50  |
| 2003                      | Spring | −0.22    | −0.44  | −0.71*    | −1.57* | −0.89*  | −1.75* | −0.13  | −0.65* |
|                           | Summer | −0.67*   | −0.96* | −0.55*    | −1.19* | −1.53*  | −1.84* | −1.32* | −1.32* |
| 2004                      | Spring | 0.91     | 1.05   | −0.34     | 0.90   | 1.54    | 1.37   | 0.14   | 1.02   |
|                           | Summer | −0.62*   | −1.10* | −0.58*    | −1.12* | −1.13*  | −2.13* | −1.74* | −1.30* |

bution, with a mean of 0 and a standard deviation of 1. The values are standardised and normalised with the following formula:

$$SPI = \frac{P_i - P_m}{S_p}$$

where  $P_i$  is the accumulated precipitation of each year ( $i$ ),  $P_m$  is the mean annual precipitation of the period analysed, and  $S_p$  is the standard deviation of the series of annual precipitations of the period under analysis.

A total of 38 rainfall stations located in the analysed basins were used to calculate the relative value of the precipitation (SPI), in each drainage area where the sampling sites were located. The aggregated rainfall data on the drainage basin of each sampling station were obtained from the Meteorological Service of Catalonia (SMC), and the SPI data were provided by the Catalan Water Agency (Table 3). Values were calculated independently for each basin. In this way, it may occur that in the same period one basin can be classified as dry while a neighbouring basin can be considered as wet. The SPI value was calculated monthly for a total period from 1996 to 2004. Data includes wet and dry episodes that can be easily discriminated using SPI. The spring season was selected as dry when the months of March, April and May had SPI values lower than  $-0.5$ , while for the selection of dry or wet summers the months of June, July and August were used. A three-month SPI analysis reflects short- to medium-term soil moisture characteristics, and it can give an indication of available moisture conditions at the beginning of the growing season and may be used as a surrogate of river flow.

#### 2.4. Analysis of differences between river types, between seasons and between dry and wet periods

River type comparisons and temporal variability between seasons (spring and summer) and between hydrological conditions (dry and wet periods) were analysed using the macroinvertebrate community assemblages at family level. Differences between samples collected in spring (from April to May) and summer (from July to August) were analysed, whereas dry and wet periods within each season were classified using the SPI values calculated in the different analysed catchments (Table 3). Also, four biological quality metrics commonly used in the Spanish Mediterranean rivers: IBMWP (Iberian Biological Monitoring Working Party), IASPT (Iberian Average Score Per Taxon) (Alba-Tercedor et al., 2004), taxon richness at family level, and the number of Ephemeroptera, Plecoptera and Trichoptera (EPT) at family level, were compared

among river types and between hydrological conditions (wet and dry periods). For metric comparisons only spring samples were used, according to the commitments agreed in the Med-GIG (Mediterranean Geographical Intercalibration Group) (European Commission, 2007). Summer data are considered more variable from year to year compared to data collected in spring and, additionally, some sites may be totally dry in summer. Thus, seasonal differences were not analysed using the biological quality metrics.

Data on macroinvertebrates of the selected reference sites were compiled from the ECOSTRIMED project database ([www.diba.cat/parcsn/qualitarius](http://www.diba.cat/parcsn/qualitarius)) ([www.ecostrimed.net](http://www.ecostrimed.net)). This is a long-term monitoring programme carried out by the Freshwater Ecology and Management (F.E.M.) research group (<http://www.pcb.ub.es/homepcb/live/es/p2907.asp>) in the Mediterranean Catalan Basins, which enabled the analysis of nine consecutive years, from 1996 to 2004, using a homogeneous data collection procedure. All samples were collected following a semi-quantitative sampling protocol (Prat et al., 2000; Jáimez-Cuellar et al., 2004). Data were gathered using four abundance ranges for each taxon (1: 1–3 individuals per sample; 2: 4–10 individuals per sample; 3: 11–100 individuals per sample; and 4: more than 100 individuals per sample). At least 200 individuals per sample were identified. Dry riverbeds or those without flow between the remaining pools (when present) were not considered in this analysis, in order to reject low abundances due to extreme dry conditions.

Rare families of macroinvertebrates (occurrence lower than 4%) were removed from the data set because such data usually can obscure patterns in classification analysis (Gauch, 1982). Differences in macroinvertebrate communities between river types were tested using an ANOSIM analysis (Clarke, 1993) based on a Bray–Curtis similarity index. An ANOSIM analysis was also performed to search for seasonal differences, and those between dry and wet periods according to the macroinvertebrate assemblages. Afterwards, an IndVal analysis (Dufrene and Legendre, 1997) was carried out in order to identify indicative families of seasons (spring and summer), and between dry and wet periods according to SPI values (see Table 3). Also, the four biological quality metrics used in the Spanish Mediterranean rivers, the IBMWP index and the IASPT index (Alba-Tercedor et al., 2004), the taxonomic richness at family level (number of total macroinvertebrate families per sample), and the EPT (number of families of Ephemeroptera, Plecoptera and Trichoptera per sample), were compared between each river type and between dry and wet periods using only spring samples as explained above.



**Table 4**  
ANOSIM results comparing macroinvertebrate communities at family level gathered on reference sites from the five river types defined in Mediterranean Catalan basins. The number of samples analysed (*N*), the *R*-value of the pair-wise comparisons, and their significance level are shown between pairs of river type comparisons.

| Comparisons between river types | Number of cases ( <i>N</i> ) | Pair-wise comparisons between river types ( <i>R</i> ) | Significance level |
|---------------------------------|------------------------------|--------------------------------------------------------|--------------------|
| 1b–2a                           | 78                           | 0.546                                                  | 0.001              |
| 1b–2b                           | 45                           | –0.016                                                 | 0.549              |
| 1b–3a                           | 69                           | 0.876                                                  | 0.001              |
| 1b–3c                           | 39                           | 0.290                                                  | 0.001              |
| 2a–2b                           | 83                           | 0.145                                                  | 0.005              |
| 2a–3a                           | 107                          | 0.378                                                  | 0.001              |
| 2a–3c                           | 77                           | 0.414                                                  | 0.001              |
| 2b–3a                           | 74                           | 0.160                                                  | 0.004              |
| 2b–3c                           | 44                           | 0.781                                                  | 0.001              |
| 3a–3c                           | 68                           | 0.057                                                  | 0.207              |

The reference conditions for each biological quality metric were calculated using the median of the values obtained in the reference sites selected for each river type, whereas the class boundary between good and very good quality was determined by the 25th percentile of reference values (Pollard and van de Bund, 2005; Erba et al., 2009). The reference values of each river type, between seasons, and values of dry and wet spring periods, were grouped when the differences between them did not exceed 5%. The 5% criterion was recommended in the harmonisation procedure of biological quality elements between EU Member States in order to group reference conditions and class boundary values (Pollard and van de Bund, 2005). Once the data sets were grouped, the differences between the aggregated groups were analysed using a non-parametric variance analysis, the Kruskal–Wallis test (Kruskal, 1964). Comparisons were performed using the SPSS v.11 software.

### 3. Results

#### 3.1. Selection of reference sites

Using the criteria shown in Table 2, 29 reference sites were selected out of 184 sampling sites analysed (Table 1). This represents 16% of the sample sites considered. In these reference sites, 171 samples were available from the ECOSTRIMED database in the period from 1996 to 2004, from which 88 samples were collected in dry periods, whereas 83 correspond to wet periods according to the SPI criteria used. From the 10 river types defined in the Mediterranean Catalan basins (Munné and Prat, 2004), only 5 of them (1b, 2a, 2b, 3a, and 3c) were found with sites that meet the reference criteria defined in Table 2. Thus reference sites were selected in 50% of sites analysed in river type 1b, in 33% of sites in river type 2a, in 38% of sites in type 2b, in 9% of sites in type 3a, and in 14% of sites in type 3c. Reference sites were mainly located in medium-sized rivers with a continuous flow regime located in headwaters. In the remaining river types (mainly located in the middle and lower part of the catchment areas) it was not possible to identify any reference

site under the selection criteria used. These rivers and streams are located in agricultural or dense urban areas, and have been submitted to intense human pressures for centuries ago. For this reason, it was not possible to select a representative number of reference sites in these river types. Besides, for most river types, the number of samples collected in summer was small, due to the fact that streams were completely dry or lacking continuous flow, especially in river type 3a.

#### 3.2. Macroinvertebrate community differences between river types

From a total of 118 families of macroinvertebrate found, 85 were used while 33 were removed from the analysis because they were poorly represented (<4% of the samples). The results of the ANOSIM analysis between river types using macroinvertebrate similarities at family level, and for a total data series of nine consecutive years (1996–2004), revealed no significant differences between rivers located in calcareous wet-mountains (type 1b), and rivers located in the calcareous middle Mediterranean-mountains (type 2b) ( $R = -0.016$ ;  $p = 0.549$ ), both with a continuous flow regime and located in the same geologic area (calcareous) (Table 4). At the same time, these two river types show significant differences with the rest of the river types, although between the types 2a and 2b, both located in the Mediterranean mid-mountain but in different geological zones (siliceous vs. calcareous), the difference is very small ( $R = 0.145$ ;  $p = 0.005$ ). The 3a and 3c river types do not display significant differences between them in the composition of macroinvertebrate communities at family level either ( $R = 0.057$ ;  $p = 0.207$ ). Both river types, the 3a and the 3c, are located in a calcareous zone and are characterized by a high flow variability. The rest of comparisons between pairs of river types analysed show significant differences between them. Higher ( $R > 0.3$ ) and significant ( $p < 0.005$ ) differences between macroinvertebrate assemblages at family level are found between rivers with a permanent and continuous flow regime (river types 1b, 2a and 2b), and temporary rivers

**Table 5**  
ANOSIM results comparing macroinvertebrate communities between spring and summer seasons and dry and wet periods using the SPI index in the five river types defined in the Mediterranean Catalan rivers. The number of samples analysed (*N*), the *R*-value of the pair-wise comparisons, and their significance level are shown.

| River type | Season | Number of cases ( <i>N</i> ) | Pair-wise comparisons between spring and summer seasons ( <i>R</i> ) | Significance level | Pair-wise comparisons between dry and wet periods of each season ( <i>R</i> ) | Significance level |
|------------|--------|------------------------------|----------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------|--------------------|
| 1b         | Spring | 11                           | 0.167                                                                | 0.014              | 0.072                                                                         | 0.212              |
|            | Summer | 9                            |                                                                      |                    | 0.094                                                                         | 0.270              |
| 2a         | Spring | 31                           | 0.122                                                                | 0.001              | 0.185                                                                         | 0.013              |
|            | Summer | 27                           |                                                                      |                    | 0.072                                                                         | 0.101              |
| 2b         | Spring | 13                           | 0.159                                                                | 0.003              | 0.191                                                                         | 0.049              |
|            | Summer | 12                           |                                                                      |                    | 0.458                                                                         | 0.003              |
| 3a         | Spring | 27                           | 0.082                                                                | 0.014              | 0.234                                                                         | 0.001              |
|            | Summer | 22                           |                                                                      |                    | –0.046                                                                        | 0.715              |
| 3c         | Spring | 10                           | 0.118                                                                | 0.041              | 0.198                                                                         | 0.048              |
|            | Summer | 9                            |                                                                      |                    | 0.650                                                                         | 0.008              |

(river types 3a and 3c). Besides, significant differences between rivers with a continuous flow regime but located in calcareous zone (river type 1b) and siliceous zone (river type 2a) are found ( $R=0.546$ ;  $p=0.001$ ). Therefore, three groups of river types can be clearly differentiated according to their macroinvertebrate composition at family level: (1) wet-mountain and mid-mountain rivers both

located in calcareous areas and characterized by a continuous flow regime (types 1b and 2b), (2) rivers located in wet mid-mountain and siliceous zones with a continuous flow regime (type 2a), and (3) temporary rivers, with a high flow regime variability, located both in semi-arid areas and karstic zones, and prone to dry-out in dry years (types 3a and 3c).

**Table 6**

Results of the IndVal analysis. Families with the most significant presence in each group are shown together with the indicator value (IndVal) ( $p$ -value < 0.05), and the average of the number of families per sample.

| River type | Season | Major families present | IndVal value                                               | Number of families               |
|------------|--------|------------------------|------------------------------------------------------------|----------------------------------|
| 1b         | Spring |                        |                                                            |                                  |
|            |        | Dry                    | Chloroperlidae<br>Nemouridae                               | 40.84<br>11.29                   |
|            | Wet    |                        | Ephemeridae<br>Perlodidae                                  | 20.36<br>13.89                   |
|            |        |                        |                                                            | 30                               |
|            | Summer |                        |                                                            |                                  |
|            |        | Dry                    | Odontoceridae<br>Perlidae                                  | 20.07<br>19.81                   |
|            | Wet    |                        | Ephemerellidae<br>Limnephilidae                            | 14.23<br>11.91                   |
|            |        |                        | Rhyacophilidae                                             | 11.58                            |
| 2a         | Spring |                        |                                                            |                                  |
|            |        | Dry                    | No families were selected                                  | 21                               |
|            | Wet    |                        | Glossosomatidae<br>Blephariceridae                         | 32.49<br>14.7                    |
|            |        |                        | Sericostomatidae                                           | 11.15                            |
|            | Summer |                        |                                                            |                                  |
|            |        | Dry                    | No families were selected                                  | 28                               |
|            | Wet    |                        | Dolichopodidae<br>Athericidae                              | 14.93<br>12.61                   |
|            |        |                        | Leptophlebiidae                                            | 9.66                             |
| 2b         | Spring |                        |                                                            |                                  |
|            |        | Dry                    | No families were selected                                  | 24                               |
|            | Wet    |                        | Empididae<br>Hydropsychidae                                | 9.27<br>6.91                     |
|            |        |                        |                                                            | 30                               |
|            | Summer |                        |                                                            |                                  |
|            |        | Dry                    | Philopotamidae<br>Limoniidae                               | 18.61<br>9.02                    |
|            | Wet    |                        | Dryopidae<br>Corixidae                                     | 42.22<br>14.15                   |
|            |        |                        | Leptoceridae<br>Leuctridae                                 | 13.27<br>11.79                   |
| 3a         | Spring |                        |                                                            |                                  |
|            |        | Dry                    | Planorbidae                                                | 11.97                            |
|            | Wet    |                        | Hydrobiidae<br>Calopterygidae                              | 8.78<br>5.83                     |
|            |        |                        |                                                            | 34                               |
|            | Summer |                        |                                                            |                                  |
|            |        | Dry                    | No families were selected                                  | 25                               |
|            | Wet    |                        | Gyrinidae<br>Hydrophilidae                                 | 13.68<br>9.68                    |
|            |        |                        |                                                            | 33                               |
| 3c         | Spring |                        |                                                            |                                  |
|            |        | Dry                    | Helodidae<br>Ceratopogonidae                               | 11.08<br>10.64                   |
|            | Wet    |                        | Pyrallidae<br>Physidae<br>Sphaeriidae<br>Cordulegasteridae | 29.47<br>27.08<br>26.25<br>26.08 |
|            |        |                        | Psychodidae<br>Psychomyiidae                               | 18.72<br>18.44                   |
|            |        |                        |                                                            |                                  |
|            |        |                        |                                                            |                                  |
|            |        |                        |                                                            |                                  |
|            |        |                        |                                                            |                                  |
|            | Summer |                        |                                                            |                                  |
|            |        | Dry                    | Aeshnidae<br>Lestidae                                      | 12.49<br>7.33                    |
|            | Wet    |                        | Anthomyidae<br>Dixidae<br>Gammaridae                       | 25.02<br>22.51<br>17.98          |
|            |        |                        | Lymnaeidae<br>Hydrometridae<br>Hydroptilidae               | 15.83<br>14.6<br>12.55           |
|            |        |                        | Coenagrionidae<br>Libellulidae                             | 12.22<br>11.71                   |
|            |        |                        |                                                            |                                  |
|            |        |                        |                                                            |                                  |
|            |        |                        |                                                            |                                  |
|            |        |                        |                                                            |                                  |
|            |        |                        |                                                            |                                  |

### 3.3. Differences in macroinvertebrate communities between seasons and between dry and wet periods

Differences are significant ( $p < 0.05$ ) between macroinvertebrate communities at family level comparing samples collected in spring and summer in the five river types analysed although these differences are not very high ( $R < 0.2$ ). Differences are slightly higher and more significant ( $p < 0.01$ ) in permanent rivers (river types 1b, 2a and 2b), than in temporary streams (types 3a and 3c) (Table 5). However, macroinvertebrate community differences are higher when dry and wet periods (differentiated using SPI values) are compared. In these cases, the  $R$  values are higher ( $> 0.4$ ), and significant differences ( $p < 0.05$ ) exist between dry and wet periods of summers in both siliceous wet mountain rivers (type 2a), and in temporary rivers located in karstic zones (type 3c). Differences are also evident, although with a lower coefficient ( $R < 0.4$ ), but with a significant value ( $p < 0.05$ ), between dry and wet periods of springs in temporary rivers located in semi-arid zones (types 3a and 3c), and in rivers with a continuous flow regime located in both siliceous and calcareous zones (types 2a and 2b) in the same period. Conversely, macroinvertebrate communities at family level do not reveal significant differences ( $p < 0.05$ ) between dry and wet periods in calcareous wet mountain rivers in either spring or summer, nor in summers of siliceous wet mountain rivers and summers of Mediterranean temporary rivers (Table 5).

The IndVal analysis performed shows the families of macroinvertebrate characteristics of each different season (spring and summer) and hydrological periods (dry and wet) (Table 6). The differences between dry and wet periods are clear from the average number of macroinvertebrate families in each sample, being lower in the dry hydrological period, whereas using spring and summer seasons this value is similar (Table 6). Thus, the hydrological period within each season characterizes the difference between communities better than just the season. Note that for river types 1b and 2a (streams with a permanent flow located in mountain areas)

**Table 7**

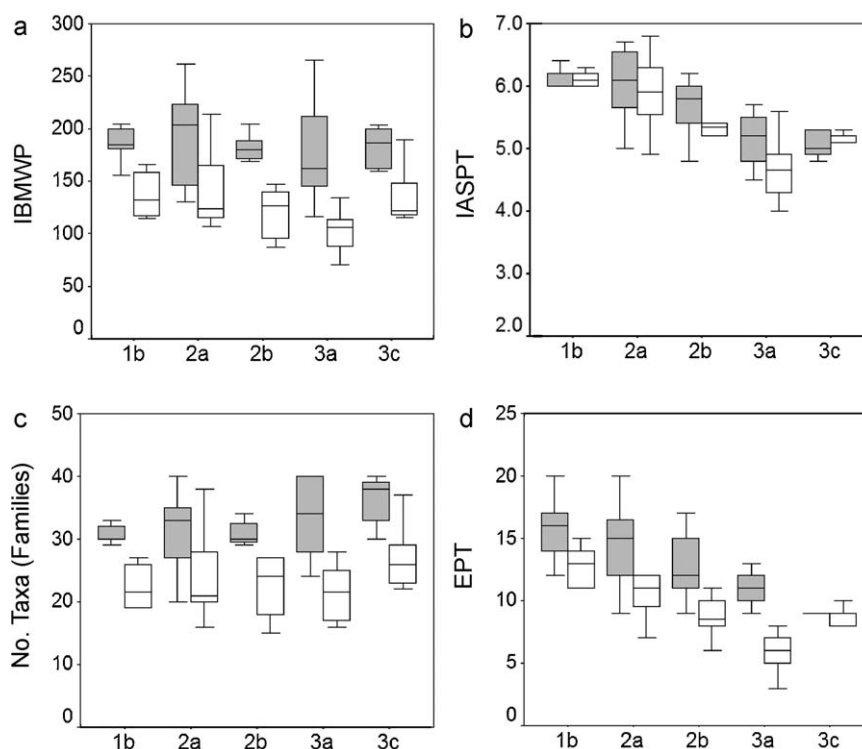
Kolmogorov–Smirnov results comparing wet and dry periods for four water quality metrics (IBMWP and the IASPT indices, the total number of taxa and the EPT taxa).

| Biological index | Z Kolmogorov–Smirnov | p      |
|------------------|----------------------|--------|
| IBMWP            | 3.296                | <0.001 |
| IASPT            | 0.904                | 0.387  |
| No. taxa         | 3.057                | <0.001 |
| EPT              | 2.031                | 0.001  |

Ephemeroptera, Plecoptera and Trichoptera orders are selected as the indicator taxa by the IndVal analysis, while for temporary streams (types 3a and 3c) even in wet periods, Coleoptera, Heteroptera, Odonata and some Diptera are the major indicative selected taxa. Besides, the total number of taxa are not so different for seasons when the hydrological period is the same (wet or dry), whereas taxon richness at family level significantly decrease from wet to dry periods in the same river type and season.

### 3.4. Differences in biological quality metrics between river types and hydrological periods

The analysis of interannual changes of four biological quality metrics reveal clear differences between median values obtained from dry and wet periods of three of them (taxon richness, EPT values and IBMWP biological indices), while the IASPT index does not show significant differences (Fig. 2, Table 7). Thus, for all river types, the median taxonomic richness in wet periods is 32 macroinvertebrate families per sample, while in dry periods this value falls to 22. Therefore, hydrological stress in the spring can be considered as the most relevant factor for taxonomic richness values, similar to the IndVal analysis results previously obtained (Table 6). The difference in the number of families present in dry and wet periods also affects the IBMWP and the EPT values, while the IASPT index, based on the relative presence of sensitive taxa, is not significantly influenced by hydrological changes (Fig. 2).



**Fig. 2.** Box-plots that show the median (in black), and 25th and 75th percentiles of reference values using the IBMWP (a), the IASPT (b), the taxa richness (c), and the number of EPT taxa (Ephemeroptera, Plecoptera and Trichoptera) (d), for each river type analysed in this paper. Wet periods are shown in grey, and dry periods in white.



**Table 8**

Mean values of the biological quality indices IBMWP and IASPT, total number of taxa and the EPT taxa between river types. The reference values (Refer.), and the high/good quality boundary (H/G) using the absolute metric value and the Ecological Quality Ratio (EQR) are shown. All values are calculated using spring samples.

| River types | No. of cases | IBMWP |     |         | IASPT |      |         | No. taxa |     |         | EPT   |     |         |
|-------------|--------------|-------|-----|---------|-------|------|---------|----------|-----|---------|-------|-----|---------|
|             |              | Refer | H/G | EQR H/G | Refer | H/G  | EQR H/G | Refer    | H/G | EQR H/G | Refer | H/G | EQR H/G |
| Wet period  |              |       |     |         |       |      |         |          |     |         |       |     |         |
| 1b          | 5            |       |     |         |       |      |         |          |     |         |       |     |         |
| 2a          | 19           |       |     |         | 6.00  | 5.50 | 0.92    |          |     |         | 15    | 12  | 0.77    |
| 2b          | 7            | 181   | 156 | 0.86    |       |      |         | 32       | 29  | 0.91    |       |     |         |
| 3a          | 13           |       |     |         | 5.20  | 4.83 | 0.93    |          |     |         | 10    | 9   | 0.90    |
| 3c          | 5            |       |     |         |       |      |         |          |     |         |       |     |         |
| Dry period  |              |       |     |         |       |      |         |          |     |         |       |     |         |
| 1b          | 6            |       |     |         |       |      |         | 22       | 20  | 0.89    |       |     |         |
| 2a          | 12           |       |     |         | 6.00  | 5.50 | 0.92    |          |     |         | 13    | 11  | 0.87    |
| 2b          | 6            | 125   | 116 | 0.93    |       |      |         |          |     |         |       |     |         |
| 3c          | 14           |       |     |         | 5.20  | 5.15 | 0.99    |          |     |         | 10    | 8   | 0.80    |
| 3a          | 5            | 106   | 89  | 0.84    | 4.65  | 4.38 | 0.94    |          |     |         | 6     | 5   | 0.83    |

Table 8 shows the aggregated values of the four metrics studied for each river type and hydrological period in the spring samples. As stated in the methods section, values were grouped for river types that do not show significant differences ( $p < 0.05$ ) according to the Kruskal–Wallis non-parametric variance analysis. The mean values of the IBMWP index, the total number of taxa, and EPT metric are different between dry and wet periods in spring samples (Table 8), while these differences are not relevant for the IASPT index except for temporary streams (3a river type). On the other hand, during each hydrological periods, metrics have different behaviour according to the river type. While the total number of taxa does not change between types, EPT is always higher in types 1b and lower in types 3a and 3c. For IBMWP, no differences between river types were found in wet periods and only type 3a had slightly lower median values in the dry period. For the IASPT index, although there are no significant differences between dry and wet periods (as stated above), it is possible to clearly differentiate between river types: while for medium and high-mountain rivers with a continuous flow regime, regardless of geological type (types 1b, 2a and 2b), the reference values are around 6, for rivers with temporary flow regime the reference value of the IASPT index is around 5.2, falling to 4.65 in dry conditions in type 3a.

#### 4. Discussion

Following implementation of the WFD, the biological quality assessment in European rivers has to be based on reference conditions calculated for each different river type at basin level. The WFD determines that the definition of river types has to be performed according to the environmental descriptors, and subsequently the assessment of reference conditions must be considered for each river type (European Commission, 2003; Nijboer et al., 2004; Noges et al., 2009). Reference sites should be established and their samples used to calculate reference values for each biological metrics. At least, several reference sites should be available for each type and the variability of the different metric values should be minimal in order to ensure there are no false positive and negative values of biological quality.

Nevertheless, it is well known that biological communities have a high temporal variability especially in rivers with large changes in hydrological conditions such as Mediterranean or other temporary rivers that may dry out (e.g., Gasith and Resh, 1999; Williams, 2006; Bonada et al., 2007a,b). The relationship between hydrological variability and biological communities has been widely studied (e.g., Blinn et al., 1995; Lake, 2003; Davey and Kelly, 2007; Boulton and Lake, 2008), including their effects on the riparian habitat structure (Poff and Ward, 1989; Poff, 1996), and on river quality assessment (Rose et al., 2008; Death et al., 2009). Many organisms inhabiting

such streams present adaptations to extreme hydrological changes (Closs and Lake, 1994; Williams, 2006; Bonada et al., 2007b), to cope with the effects of habitat changes over time (Blinn et al., 1995; Gallardo-Mayenco et al., 1995). This is clearly reflected in the biological traits of the species. While most macroinvertebrates do not present specific traits in continuous flow regimes, in temporary rivers, and especially in dry periods, species need to adapt to the harsh changes produced by lenitic conditions (e.g., decreasing dissolved oxygen and high temperatures), and therefore organisms should have biological traits (e.g., from active dispersion to forms of resistance) that enable them to persist when the river dries up (Boulton and Lake, 1992; Williams, 2006; Bonada et al., 2007a). Thus, the macroinvertebrate assemblage structure changes over time according to the hydrological conditions (from high to low water flow) in temporary streams (e.g., Boulton and Lake, 1992; Boulton, 2003; Sangiorgio et al., 2007), and, as a result, the values of different biological metrics in reference conditions may be different according to the hydrological conditions in the months prior to the sampling (Buffagni et al., 2009). Nevertheless, temporal variability is not usually considered for quality assessment using macroinvertebrates in Mediterranean streams. In most papers the hydrological period is mainly seasonally considered (i.e. spring vs. summer) and not by using hydrology, whose characteristics can vary greatly within a season in different years according to the climate conditions.

Our analysis shows that macroinvertebrate communities vary significantly between hydrological conditions. Dry and wet periods differ more (within each season) than seasonal (spring vs. summer) changes. At the same time, several differences have been found among the five river types examined. From these five different river types studied in the Mediterranean Catalan area (NE Spain), a clear differentiation on community composition at family level has been appreciated, clustering the rivers in three different groups: (1) rivers with a continuous flow regime located in siliceous zones; (2) rivers with a continuous flow regime located in calcareous zones; and (3) temporary rivers regardless from the geology. These results agree with those found in a similar Mediterranean areas for Sánchez-Montoya et al. (2007), or other European areas (e.g., Verdonshot, 2006) where the environmental typology does not match the biological communities. The geology of the basin can be considered as a key element that determines the presence of certain macroinvertebrate families in rivers with continuous flow regime, and this has also been detected by other studies made in the same region using Chironomidae (Puntí et al., 2007). Moreover, temporary rivers appear in all these studies as a heterogeneous group, without a unique typological aggregation but displaying notable dispersion between sites and samples, with a more variable composition than any other group as stated in Sánchez-Montoya

et al. (2007). This phenomenon is accentuated in samples taken in summer. In temporary rivers, part of the same river channel may be subject to different habitat conditions due to a high variability of flow regime patterns, and the spatial comparisons among sites may be worthless. Similar results have been found in other temporary rivers, mainly located in dryland areas (e.g., Sheldon, 2005). They clearly show that not all the environmental types defined using the WFD approach may be differentiated by their community composition, and biological data should further be considered in order to define river types and calculate their reference conditions.

Moreover, our results explain that the characteristics of hydrological periods within each season (dry vs. wet) characterize the differences between communities better than just the season. The composition of macroinvertebrate communities in temporary rivers is characterized by the presence of families of Odonata, Coleoptera and Heteroptera (OCH) orders rather than the Ephemeroptera, Plecoptera and Trichoptera (EPT). The OCH share biological traits that allow them to survive in non-flowing streams (only pools) or migrate when the stream totally desiccates, compared with many species inhabiting permanent rivers (Williams, 1996; Magoulick and Kobza, 2003). Those taxa are present in each river in accordance with local hydrological conditions, which usually coincide with the season (summer season is normally dry) but may display high interannual changes (Buffagni et al., 2009). Thus, the difference between permanent and temporary streams can be clearly found using the EPT+OCH/EPT ratio proposed by Rieradevall et al. (1999) and Bonada et al. (2006). Low values for this index are found in dry periods compared to wet ones, because pool species (OCH) increased their importance in dry periods. As a consequence macroinvertebrate assemblages, richness and abundance of tolerant taxa, show significant differences between hydrological periods in the same river type as demonstrated in this paper (Tables 6 and 8). Therefore, specific hydrological conditions should additionally be considered in order to better calculate biological reference conditions, and to properly apply biological metrics used to establish the ecological status of rivers, especially in temporary ones. Our results are very relevant for managing purposes, because significant differences in two quality metrics, widely used in biological surveys of rivers, the number of taxa, and the IBMWP index (Alba-Tercedor et al., 2004), exist for the different hydrological periods but not for seasons, nor for the river types analysed. Thus, the use of such metrics in calculating the median value to define the reference conditions without considering the hydrological period may yield misleading results, because modify the values of quality class boundaries. In the same way, the use of the 25th percentile to establish the moderate-good quality boundary (Pollard and van de Bund, 2005) can overestimate or underestimate the quality classification and influence the target for management purposes. The use a unique value for both dry and wet periods is incorrect in order to define reference conditions for those metrics. Similar results have been found by calculating river health during drought events in other temporary rivers (e.g., Rose et al., 2008), because values of metrics in reference condition streams decrease when flow resumes. Biological communities change according to flow regime, especially in temporary and Mediterranean rivers, and this must be considered in order to establish reference conditions for biological metrics sensitive to these changes (e.g., the IBMWP index).

Conversely, other quality metrics, such as the IASPT index analysed in this paper, are more type specific, and do not show significant differences between hydrological periods. As we explained before, lenitic taxa (e.g., Odonata, Coleoptera and Heteroptera–OCH) are more tolerant to hydrological changes and replaced sensitive and flow-requiring taxa (Ephemeroptera, Plecoptera and Trichoptera–EPT) during dry periods. Most of the EPT taxa (present in permanent streams) are not tolerant taxa and they

have high score values in the IBMWP (and other similar) system, while the OCH taxa, usually have lower quality scores. Moreover, the IBMWP index uses absolute values adding score values for each taxon, which may sometimes produce differences in the total score between hydrological periods except when the number of OCH is very high and compensates the loss of EPT, which implies higher taxon richness (e.g., Bonada et al., 2006). On the other hand, relative values (e.g., IASPT) enable data comparison between hydrological periods. Relative values used by IASPT index, that are not habitat dependant, is more suitable for use in rivers with a high temporal variability, whose habitat changes according to the flow regime. The IASPT index analysed does not show great differences between hydrological periods but has significant differences between river types, especially between permanent and temporary streams.

Biological quality metrics need to be previously analysed in order to know their temporal or spatial behaviour, and to select the most suitable in each case. For Mediterranean areas, and especially for temporary streams, relative values seem more appropriate than biological indices or taxon richness if the hydrological period is not considered. In this case, the IASPT, and also multimetric indices that use this metric in their formulation, such as the ICM-Star (Buffagni et al., 2006), the IMMi-L and the IMMi-L indices (Munné and Prat, 2009) seem to be more appropriate. On the other hand, if the hydrological period is previously considered, the IBMWP index can be used if the particular values of this metric are calculated in reference conditions for dry and wet periods. The use of the dry–wet period classification according to the climate characteristics (precipitation as surrogate of hydrological conditions) results in a more accurate application of the WFD in Mediterranean rivers, where large hydrological changes may occur. Thus, median values will not be overestimated in dry periods and underestimated in wet periods, and EQR values will be less variable, a fact that has so far been observed in studies of temporary streams. We also emphasize herein the need of long-term monitoring programmes.

## 5. Conclusions

We suggest to characterize the hydrological characteristics (dry or wet), prior to assessing the biological quality, especially in temporary streams, as is the usual case in many small Mediterranean streams. The SPI index based on rainfall in the three months prior to sampling can be used as a surrogate to previously classify the hydrological periods of each sampling period (when continuous flow monitoring is not available, which is the case of most small streams), and afterwards select the adequate reference conditions for biological metrics. On the other hand, differences in community composition and values of biological quality metrics, according to different environmental river types, have been previously reported (e.g., Sánchez-Montoya et al., 2007), and even recognized in the intercalibration exercise carried out in the Mediterranean GIG (Heiskanen et al., 2004). For this reason, care should be taken when medians and percentiles of the biological quality values are used to define quality class boundaries. This can be solved using the approach taken in this paper: classifying samples according to hydrological conditions: dry or wet periods (e.g., using the SPI index), and afterwards calculating the EQR for biological metrics using suitable reference conditions for each hydrological period. Nowadays, several countries in Europe are carrying out an intercalibration process in order to harmonise biological indices and their quality class boundaries (Heiskanen et al., 2004; Noges et al., 2009). The intercalibration procedure, the quality metrics tested, and the reference conditions calculated are based on spatial differences, on river types, according to WFD requirements. However, as we have clearly demonstrated in this paper, some biological quality indices and the reference conditions subsequently derived can show a high

variability in the same river type when different hydrological periods are considered. This phenomenon is much more accentuated in temporary rivers, where changes in hydromorphology due to flow regime conditions are highly variable over years and between years. Implications of future climate changes may be envisaged from our data, but this needs a more accurate use of data to make predictions for the future and is beyond the scope of this paper.

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